PUNJAB
FLOODPLAIN FOREST RESTORATION

Sector: Agriculture
Hazard adressed: Flood
Targeted assets: Rangeland

Description: This measure involves restoration of degraded floodplains through plantation of native tree species, grasses, and shrubs to improve ecosystem functions and natural flood retention capacity. The intervention enhances riverbank stabilization, reduces soil erosion, improves groundwater recharge, restores biodiversity, and strengthens resilience of communities and ecosystems against flooding and climate variability.

Priority Districts: Bhakkar, Chakwal, Chiniot, Gujranwala, Khushab, Lahore, Mianwali, Muzaffargarh, Nankana Sahib, Okara, Pakpattan, Rajanpur, Rawalpindi, Sahiwal, Sheikhupura, and Sialkot.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Hectare (Ha) of restored floodplain area.

Scale assumptions: Applicable in degraded floodplain and river corridor areas vulnerable to erosion, flooding, vegetation loss, and ecosystem degradation.

Key design parameters: Use of native flood- and drought-tolerant tree/shrub species, soil stabilization measures, ecological buffer zones, community-led plantation activities, and controlled grazing management.

Time horizon: 2030 - 2037; ecological and hydrological benefits continue over long-term periods.

IMPACT

Risk reduction level: Moderate to High.

Time to effectiveness: Medium- to long-term following vegetation establishment and ecosystem restoration.

Expected impact: Reduced flood runoff and soil erosion, improved riverbank stability, enhanced groundwater recharge, restored ecosystem functions, and strengthened resilience against flood and drought hazards.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Availability of native species, community participation, watershed management planning, and long-term maintenance support.

Required institutions: Forest Department Punjab, Irrigation Department Punjab, Environment Protection Agencies, Local Government Department, and development partners.

Barriers: Land tenure issues, grazing pressure, limited maintenance funding, climate stress affecting vegetation survival, and weak institutional coordination.

COST ASSUMPTIONS

Unit cost: PKR 38 per sapling or PKR 23,750 per hectare (~USD 85 per hectare).

Capital Expenditure (CAPEX):
USD 2.5 million per year (2030 – 2034). Includes saplings, plantation activities, site preparation, soil stabilization, fencing/protection, and initial maintenance.

Operational Expenditure (OPEX):
USD 2.1 million per year (2035 - 2037). Includes maintenance, replacement planting, monitoring, and ecosystem management activities.

CO-BENEFITS & TRADE-OFFS

Environmental: Enhanced biodiversity, carbon sequestration, watershed protection, and restoration of degraded ecosystems.

Social: Improved livelihood resilience and ecosystem services for vulnerable rural communities.

Economic: Reduced flood-related damages and improved long-term sustainability of natural resource systems.

Potential risks: Poor survival of saplings, overgrazing pressure, weak maintenance systems, and delayed ecological recovery.

BENEFICIARIES

Primary beneficiaries: Rural communities, farmers, pastoral households, forestry departments, and downstream populations vulnerable to flooding and land degradation.

Distributional aspects: Particularly beneficial for vulnerable floodplain communities exposed to recurrent flooding, erosion, and environmental degradation.

Note: This factsheet is to be used as a rapid planning and appraisal tool. All costs and impacts should be validated through field assessments, stakeholder consultation and detailed feasibility studies.